Career Launcher MBA

QA Algebra

Review Test – 02

Find the range of values of x for which the following inequality holds true

$$\frac{4x^2}{(1-\sqrt{2x+1})^2} < 2x+9?$$

(1) $x \in (-45/8, -1/2)$

(2) $x \in (-1/2, 45/8) - \{0\}$

(3) $x \in (-1/2, 45/8)$

(4) $x \in (-45/8, 45/8)$

Question – 1

How many natural number triplets (x, y, z) are possible such that when any of these numbers is added to the product of the other two, the result is 2.

(1) None

(2) 1

(3) 2

(4) Infinitely many

Question – 2

How many integral solutions for the following equation $a^2 + ab + 4a + 2b = 0$ are possible?

(1) None

(2) 3

(3) 6

(4) 12

Question – 3

Find the value of x such that f(x) will be minimum, where $f(x) = \Sigma|nx - 1|$; for n = 1 to 10.

(1) $x \in (1/5, 1/6)$

(2) $x \in (5, 6)$

(3) $x \in \{1/7\}$

(4) $x \in \{7\}$

Question – 4

Find the number of terms in the expansion of $(1 + x^3 + x^5)^{10}$.

(1) 66

(2) 51

(3) 46

(4) 45

Question – 5

A quadratic polynomial $f(x) = ax^2 + bx + c$ and $x \neq 0$ satisfies the following conditions:

1. $f(-5) = 0$
2. $f(14) = f(56)$

Find $f(0)/(f(10))$.

(1) – 5/13

(2) 5/13

(3) 15/17

(4) Cannot be determined

Question – 6

What will be the coefficient of $a^8 b^{12}$ in the expansion of $(a + b)^{21}$?

(1) Cannot be determined

(2) ${}^{21}C_8 \times {}^{13}C_{12}$

(3) $(21!)/(8! \times 12!)$

(4) None of the above

Question – 7

$[1^2/100] + [2^2/100] + [3^2/100] + [4^2/100] + \ldots + [k/100] = 45$ where k is a natural number. Find the digital sum of k.

Digital sum is the recursive one digit sum of all digits of a number. For example, digital sum of 235 = 1.

[.] denotes Greatest Integer Function/Floor Function.

(1) 4

(2) 5

(3) 6

(4) 7

Question – 8

For how many integral values of x the inequality below holds true?

$\log_{0.3}[\log_x(2x^2 - 15x - 100)] > \log_{0.3}(2)$

(1) 18

(2) 19

(3) 24

(4) Infinite

Question – 9

$A = (2 + 1)(2^2 + 1)(2^4 + 1) \ldots (2^{2048} + 1)$. The value of $(A + 1)^{1/2048}$ is

(1) 4

(2) 2048

(3) 2^{4032}

(4) 2

Question – 10

The 12 numbers a_1, a_2,....., a_{12} are in Arithmetic Progression. The sum of all these numbers is 354.

Let P = $a_2 + a_4 + ... + a_{12}$ and Q = $a_1 + a_3 + ... + a_{11}$. If the ratio P : Q is 32 : 27, the common ratio of the progression is

(1) 2

(2) 3

(3) 4

(4) 5

Question – 11

a and b are positive integers such that $a^2 + 2b = b^2 + 2a + 5$. The value of b is _____.

Question – 12

In the sequences 1, 2, 2, 4, 4, 4, 4, 8, 8, 8, 8, 8, 8, 8, 8, ... the 2016th term is 2^n. Then n = _____.

Question – 13

Diophantus's riddle is a poem that encodes a mathematical problem. In verse, it read as follows:

'Here lies Diophantus,' the wonder behold.

Through art algebraic, the stone tells how old:

'God gave him his boyhood one–sixth of his life,

One twelfth more as youth while whiskers grew rife;

And then yet one-seventh ere marriage begun;

In five years there came a bouncing new son.

Alas, the dear child of master and sage,

After attaining half the measure of his father's life chill fate took him.

After consoling his fate by the science of numbers for four years, he ended his life.'

What was sum of the ages Diophantus and his son when the latter died?

(1) 100 years (2) 122 years

(3) 124 years (4) 126 years

Question – 14

Mansi counted the total number of vehicles in a traffic jam (consisting of at least one each of motorcycle, 3 wheeler auto rickshaw and car) to be 150 and total number of wheels (not spares) to be 320. If she knew that there were more cars than auto rickshaws, then the number of vehicles having more than 2 wheels can be

(1) 11

(2) 12

(3) 13

(4) All of the above

Question – 15

$f(x) = ax^2 + bx + c$, for $a \neq 0$ is a function whose roots do not lie in the interval $(-1,1)$. Which of the following holds true?

(1) $a + c > 0$

(2) $a^2/(b + c) > 1$

(3) $(a + c)^2/b^2 > 1$

(4) $b^2/(a + c) > 1$

Question – 16

If all the roots of f(x) are real and positive integer, and $f(x) = x^{100} - px^{99} + qx^{98} + rx^{97} + \ldots + 1$ then which of the following can be a value of p?

(1) –100

(2) 1

(3) 99

(4) 100

Question – 17

What is the minimum value of $a + \frac{1}{\{b(a-b)\}}$ where a and b are positive real numbers and a > b.

(1) 2

(2) 3

(3) 4

(4) 5

Question – 18

Which of the following statements is true about the following two terms for $n > 1$?

(a) $n!$

(b) $[(n + 1)/2]^n$

(1) $n!$ is larger than $[(n + 1)/2]^n$

(2) $[(n + 1)/2]^n$ is larger than $n!$

(3) Both can be same for some value of n

(4) Nothing can be said about the above two

Question – 19

If $a_0 = 1$, $a_1 = 1$ and $a_n = a_{n-1} \times a_{n-2} + 1$ for $n > 1$ then

(1) a_{999} is odd and a_{1000} is even

(2) a_{999} is odd and a_{1000} is odd

(3) a_{999} is even and a_{1000} is even

(4) a_{999} is even and a_{1000} is odd

Question – 20

Fibonacci series is a series where $a_1 = 0$, $a_2 = 1$ and $a_{n+2} = a_{n+1} + a_n$ for $n > 0$. Which of the following is not true about this series?

(1) a_{30} is a multiple of a_2, a_3 and a_5

(2) $a_1 + a_2 + \ldots + a_{28} = a_{30} - 1$

(3) $a_1^2 + a_2^2 + \ldots + a_{29}^2 = a_{29} \times a_{30}$

(4) $a_1 + a_2 + \ldots + a_{30}$ is odd

Question – 21

If the sum of n terms of a series is $2n^2 – n$ then find its nth term.

(1) $3n – 4$

(2) $4n – 3$

(3) $2n^2 – 1$

(4) Cannot be uniquely calculated

Question – 22

$(6666666666)^2 + (8888888888) =$

(1) $(4444444444)^2$

(2) (44444444444444444444)

(3) (22222222222222222222)

(4) $(8888888888) + (44444)^2$

Question – 23

An exam named MEOW (Management Exam for Orphic Women) has 50 questions. For each question correctly answered, a candidate receives + 4 and for every wrong answer, – 2 are awarded. Unattempt questions do not carry any marks. Smarta wants to know the sum of all possible scores that can be scored in the exam which is

(1) 6713

(2) 7361

(3) 7352

(4) 7631

Question – 24

Vicky has infinite notes of custom denomination of Rs. 9 and Rs.11. Which is the highest amount he cannot form using notes of these two denominations? Rs. _____

Type 0 if answer is “Cannot be determined” else type in the answer in the blank space provided.

Question – 25

f(a, b) = average of a and b; g(a, b) = max of a and b; h(a, b) = min of a and b. Then how many of the following statements is/are true?

(i) f(g(a, b), h(a, b)) > f(a, b)

(ii) g(f(a, b), h(a, b)) < h(a, b)

(iii) f(f(f(a, b), a), b) > h(a, b)

(iv) h(f(a, b),g(a – b, a + b)) < h(a, b)

(1) None

(2) One

(3) Two

(4) Three

Question – 26

Find number of terms in the expansion of

$(a + b + c)^4(b + c + d)^3$

(1) 150

(2) 33

(3) 63

(4) None of the above

Question – 27

If $(a + b - c)/c$, $(b + c - a)/a$ and $(c + a - b)/b$ are in AP ($a, b, c \neq 0$), then which of the following is/are definitely true?

(1) $a = (b + c)/2$

(2) $b = (a + c)/2$

(3) $c = 2ab/(a + b)$

(4) $a = 2bc/(b + c)$

Question – 28

f(x) is a polynomial of degree 49 such that f(1) = 4, f(2) = 5, f(3) = 6, ... f(49) = 52. Find f(50). (k is positive constant)

(1) 53 + k49!

(2) 53!

(3) 54!

(4) 49!

Question – 29

x, y, z are distinct real numbers, not equals to 0, such that $x + \frac{1}{y} = y + \frac{1}{z} = z + \frac{1}{x}$.

The value of $x^2y^2z^2$ is _____.

Question – 30

Thanks!